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Your grade: 100%

Your latest: 100% • Your highest: 100%

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Learning Check - Interaction Terms

Review Learning Objectives

Instructions

1. Suppose we have a dataset of 100 students for which we have access to two features: $X_1 = GPA$ and $X_2 = Gender$ ($X_2 = male$ or $female$). Your job is to fit the following model to the data and report the value of β_3 (in thousands of dollars). 1 / 1 point

Help me practice

Let's chat

Suppose we use least squares regression to fit the following model:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times GPA + \hat{\beta}_2 \times Gender + \hat{\beta}_3(GPA \times Gender),$$

and obtain the values $\hat{\beta}_0 = 35, \hat{\beta}_1 = 5, \hat{\beta}_2 = -15, \hat{\beta}_3 = 5$.

Assignment Details

Due

Attempts

For what range of values of GPA do females earn more than males?

Submitted Sep 24, 1:51 AM IST

Try again

Correct

100%

Write down two linear equations: One for $\hat{Y}$ as a function of X_1 when $X_2 = 0$ and another one for $\hat{Y}$ as a function of X_1 when $X_2 = 1$. These two linear equations represent the predicted started salary for males and females, respectively. Find out the crossing point of the two lines represented by these equations. This is done by solving a simple linear equation with solution $X_1 = 3.0$. Hence, (on average) females earn more than males for $X_1 > 3.0$.

2. True or False: Since the coefficient $\hat{\beta}_3$ for the $GPA \times Gender$ interaction is large, there is statistical evidence of a strong interaction effect. 1 / 1 point

True

False

Correct

False, we would have to look at the t-statistic.

3. Suppose that, using the same data, we fit another predictor using the following model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times GPA + \hat{\beta}_2 \times Gender + \hat{\beta}_3(GPA \times Gender) + \hat{\beta}_4(GPA^2 \times Gender)$. Is the training RSS for this model lower than or equal to ($\leq$) the training RSS for the previous model $\hat{Y}$? 1 / 1 point

Yes

No

Not enough information

Correct

Since the model is more flexible, the training RSS either stays the same or is reduced.

4. Suppose that, using the same data, we fit another predictor using the following model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \times GPA + \hat{\beta}_2 \times Gender + \hat{\beta}_3(GPA \times Gender) + \hat{\beta}_4(GPA^2 \times Gender)$. Is the test RSS for this model lower than or equal to ($\leq$) the test RSS for the previous model $\hat{Y}$? 1 / 1 point

Yes

No

Not enough information

Correct

We cannot tell with the information given. Our model may be overfitting the training data.

https://www.coursera.org/learn/machine-learning-essentials/assignment-submission/4RtsJ/learning-check-interaction-terms/view-feedback

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